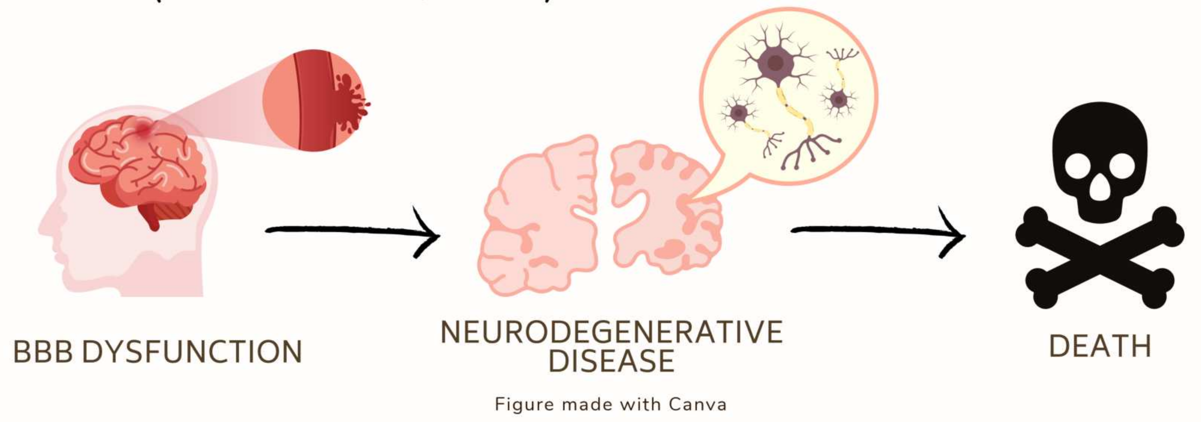
Improving the Integrity of the Blood-Brain Barrier Through Stimulation of Lactate Receptor HCAR1

Problem

The blood-brain barrier is a **semi-permeable barrier** that controls the entry of substances into the brain from the bloodstream. When it is impaired, toxic substances can enter the brain, causing **neuronal dysfunction** (Sharma et al., 2022). It has been linked to several **neurogenerative diseases**, including Alzheimer's, and can eventually lead to death (Sharma et al., 2022).



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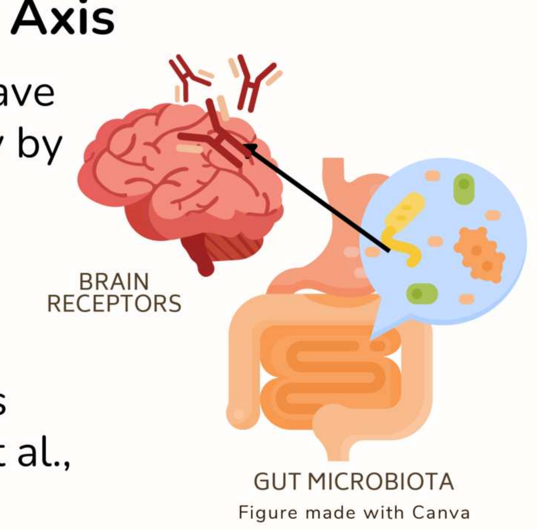
Background

Gut-Brain Axis

Several microbial metabolites have been shown to improve integrity by activating **G-protein coupled receptors** (Hoyles et al., 2018). Current research suggests that lactate is able to **reach the BBB directly from the gut**, though its effects are not yet known (Cai et al., 2022).



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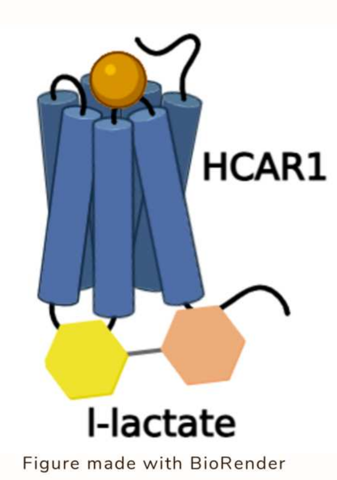
Lactate Receptor HCAR1



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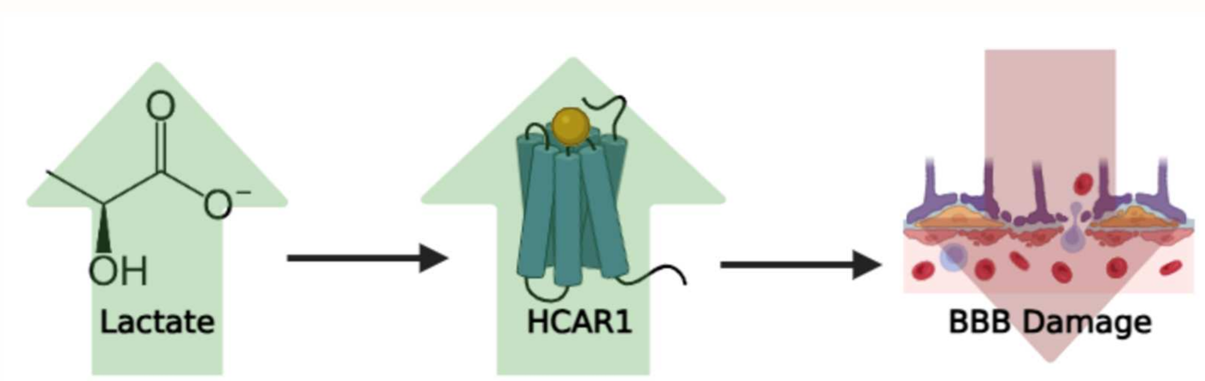
The lactate receptor HCAR1 is found at the BBB, and previous studies have shown that its activation leads to **increased mitochondrial biogenesis**, necessary for maintenance of the BBB (Boitsova et al., 2018). Furthermore, it was found that **lactate activated HCAR1** in the brain and **stimulated angiogenesis**, which is also necessary for the maintenance of the BBB (Morland et al., 2017).

Hypothesis

It was hypothesized that an **l-lactate treatment upregulated levels of the lactate receptor HCAR1**, leading to a decreased permeability of the blood-brain barrier (indicating an improved integrity).



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Independent Variable: 1 mM, 5 mM, and 20 mM l-lactate
Dependent Variable: HCAR1 expression

Methods

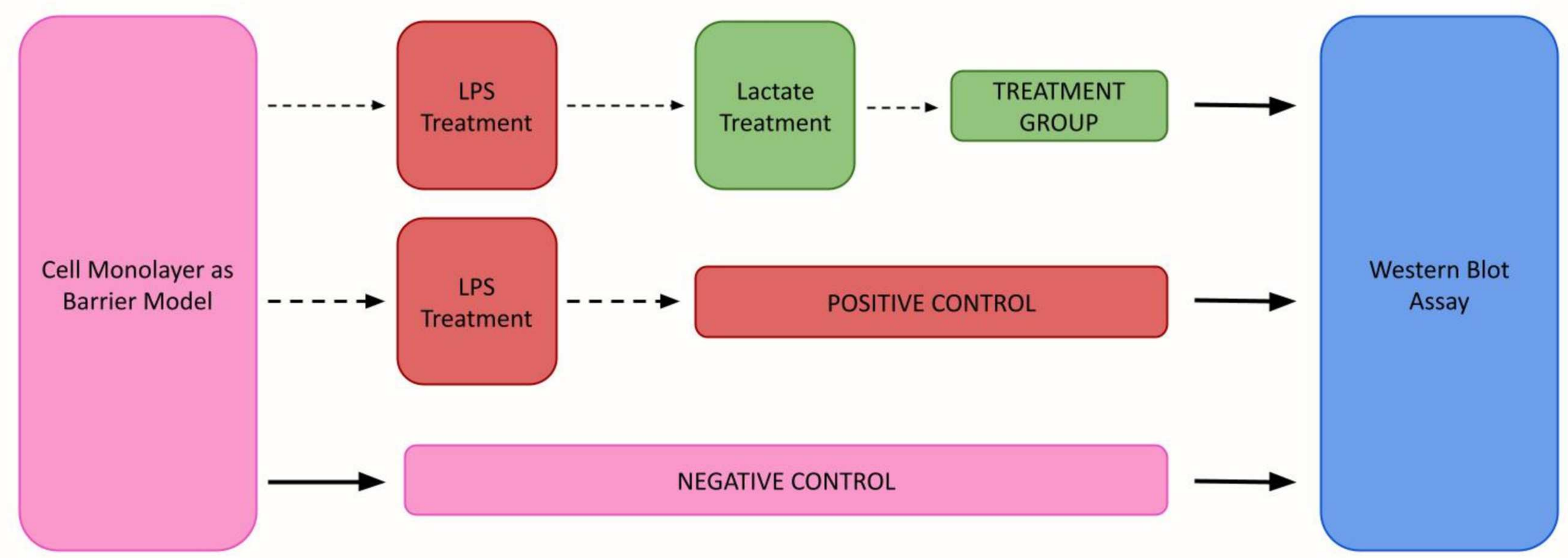

Figure 1: A flowchart representation of the procedure.

Barrier Model

- The **Caco-2 cell line** was used, as previous studies have shown that it mimics the permeability of the blood-brain barrier (Shah & Dong, 2022).
- A lipopolysaccharide (LPS) treatment was used to induce damage to the barrier model.

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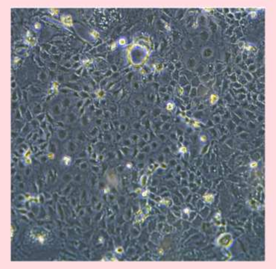


Image 1: Caco-2 cells in 6-well plate at full confluency with no treatment applied.

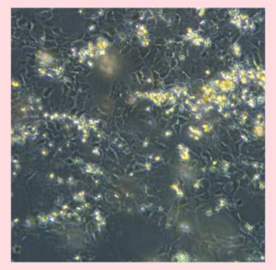
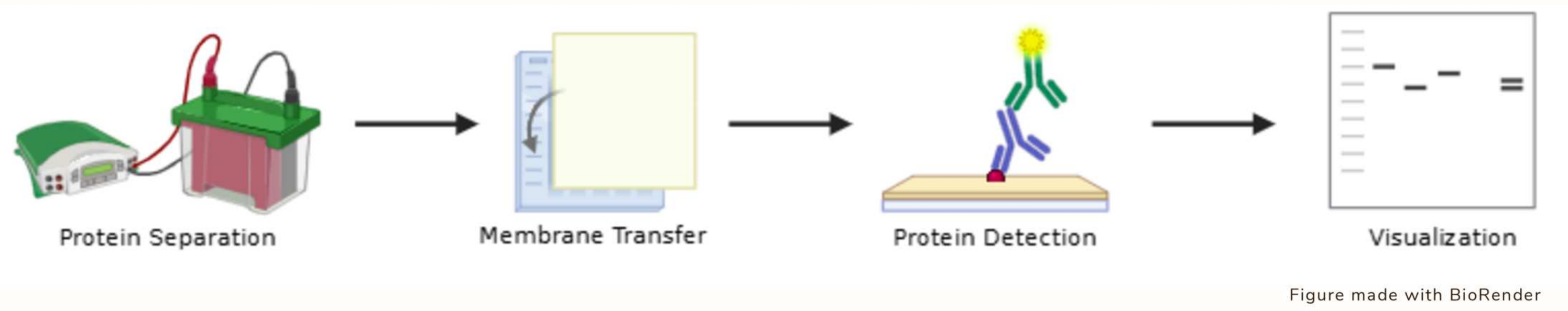


Image 2: Caco-2 cells in 6-well plate 48 hours after adding LPS treatment.

Western Blot Assay

- The **levels of HCAR1** were measured.
- **ImageJ** was used to quantify the results.



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Data

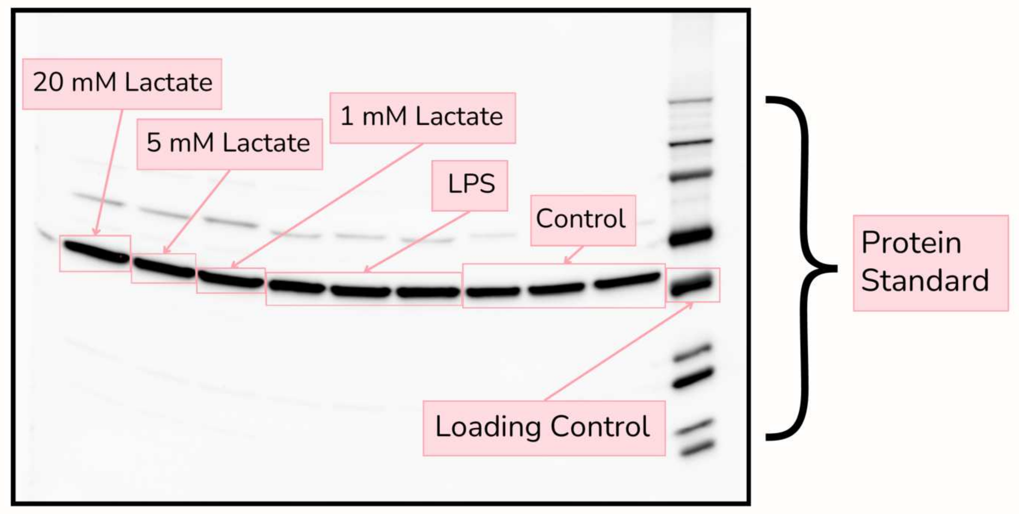

Image 3: Western blot #1

Five western blots were run, with processed data below.

	HCAR1 % of Control				
	Control	LPS	1 mM Lactate	5 mM Lactate	20 mM Lactate
Trial 1	0.768	0.790	0.867	0.881	0.860
	0.670	0.787	0.336	0.431	0.485
	0.668	0.816	0.476	0.358	0.469
Trial 2	0.401	0.479	0.382	0.437	0.443
	0.398	0.432	0.374	0.475	0.469
	0.316		0.455	0.573	0.399
Trial 3	0.396	0.380	0.444		0.547
	0.380	0.425	0.375	0.352	0.430
	0.420	0.506	0.359	0.444	0.464

Note: The blank boxes are from samples that did not image properly on the blot.

Table 1: HCAR1 % of loading control for each trial

Results

A nonparametric Kruskal-Wallis test was performed on the HCAR1 expression values; it resulted in a p value of 0.2890, indicating that there was no significant difference in HCAR1 expression between treatments.

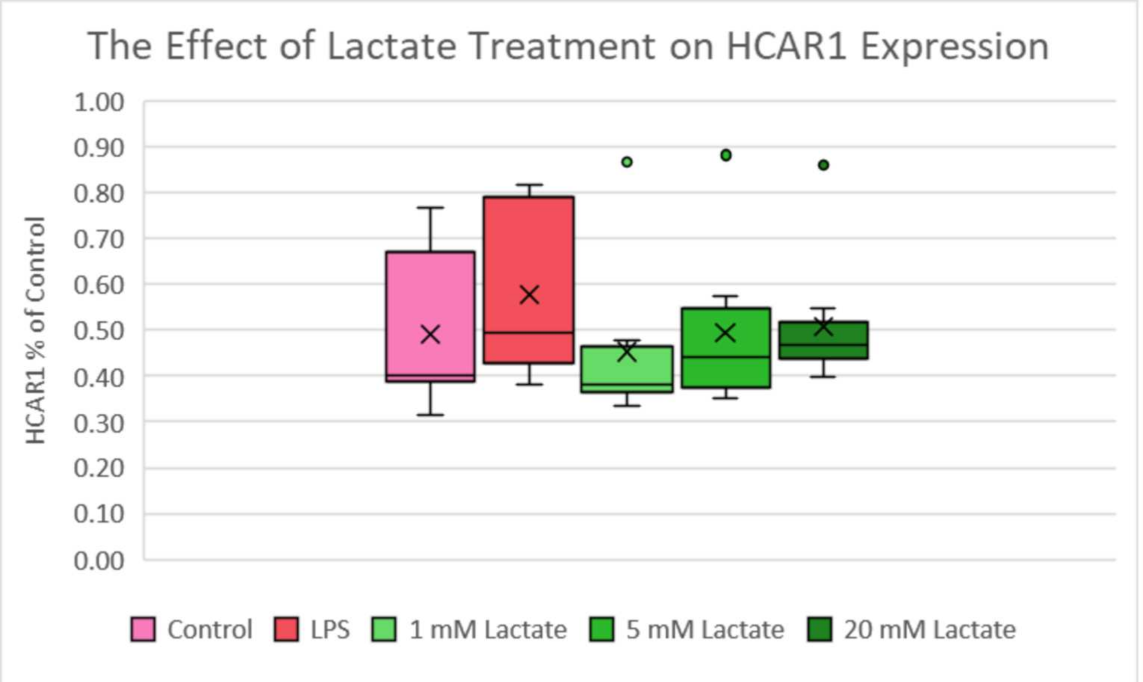

Figure 2: The levels of HCAR1 expression after treatment (n = 8-9 replicates per group)

Conclusions

The results of this experiment **did not support the hypothesis**.

- There was no significant difference in HCAR1 expression after the LPS or the l-lactate treatments, compared to the control group.
- After **LPS treatment**, there was a **slight increase in HCAR1 expression**, though not statistically significant, rather than a decrease, as expected.
- The **l-lactate treatments** did **not increase HCAR1 expression**.

Last year, it was found that **LPS significantly damaged the permeability** of the barrier model, and, regardless of concentration, **l-lactate treatment was able to improve integrity** closer to control levels.

- This suggests that l-lactate's effects on barrier permeability **may not be linked to its receptor HCAR1**.

Future Work

In the future, further research is necessary to **determine the pathways** through which lactate is able to affect the integrity of the barrier, if not expression of HCAR1. The treatments also need to be tested on **blood-brain barrier cells** and **in vivo** to make sure that the effects are mimicked there and that lactate reaches the BBB from the gut.

References

Boitsova, E. B., Morgun, A. V., Osipova, E. D., Pozhilenkova, E. A., Martinova, G. P., Frolova, O. V., Olovannikova, R. Y., Tohidpour, A., Gorina, Y. V., Panina, Y. A., & Salmina, A. B. (2018). The inhibitory effect of LPS on the expression of GPR81 lactate receptor in blood-brain barrier model in vitro. ***Journal of Neuroinflammation***, ***15***(1). https://doi.org/10.1186/s12974-018-1233-2

Cai, M., Wang, H., Song, H., Yang, R., Wang, L., Xue, X., Sun, W., & Hu, J. (2022). Lactate is answerable for brain function and treating brain diseases: Energy substrates and signal molecule. Frontiers in Nutrition, 9. https://doi.org/10.3389/fnut.2022.800901

Hoyles, L., Snelling, T., Umlai, U. K., Nicholson, J. K., Carding, S. R., Glen, R. C., & McArthur, S. (2018) Microbiome-host systems interactions: protective effects of propionate upon the blood–brain barrier. ***Microbiome 6,*** 55. https://doi.org/10.1186/s40168-018-0439-y

Morland, C., Andersson, K. A., Haugen, Ø. P., Hadzic, A., Kleppa, L., Gille, A., Rinholm, J. E., Palibrk, V., Diget, E. H., Kennedy, L. H., Stølen, T., Hennestad, E., Moldestad, O., Cai, Y., Puchades, M., Offermanns, S., Vervaeke, K., Bjørås, M., Wisløff, U., Storm-Mathisen, J., Bergersen, L. H., (2017). Exercise induces cerebral VEGF and angiogenesis via the lactate receptor HCAR1. ***Nature communications, 8***, 15557. https://doi.org/10.1038/ncomms15557

Sharma, C., Woo, H., & Kim, S. R. (2022). Addressing Blood-Brain Barrier Impairment in Alzheimer's Disease. Biomedicines, 10(4), 742. https://doi.org/10.3390/biomedicines10040742